E-COMMERECE SALES ANALYSIS

INTRODUCTION:

The goal of this project is to clean and transform raw product data to ensure accuracy, consistency, and usability. This involves handling missing values, correcting inconsistent text formats, fixing typos, removing duplicates, splitting and merging columns, applying proper number formatting, and using conditional formatting.

DATASET DESCRIPTION:

- Data entry, import, and basic organization using tables
- Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data
- Remove the column that contains irrelevant data if it exists
- Application of Excel formulas for calculations, summaries, and logical/text operations
- Use XLOOKUP/VLOOKUP functions to combine the tables.
- Use Pivot Tables for advanced summarization
- Descriptive statistics (mean, median, mode, standard deviation, etc.)
- Data visualization to create three chart types, build an interactive dashboard, and use basic PivotCharts

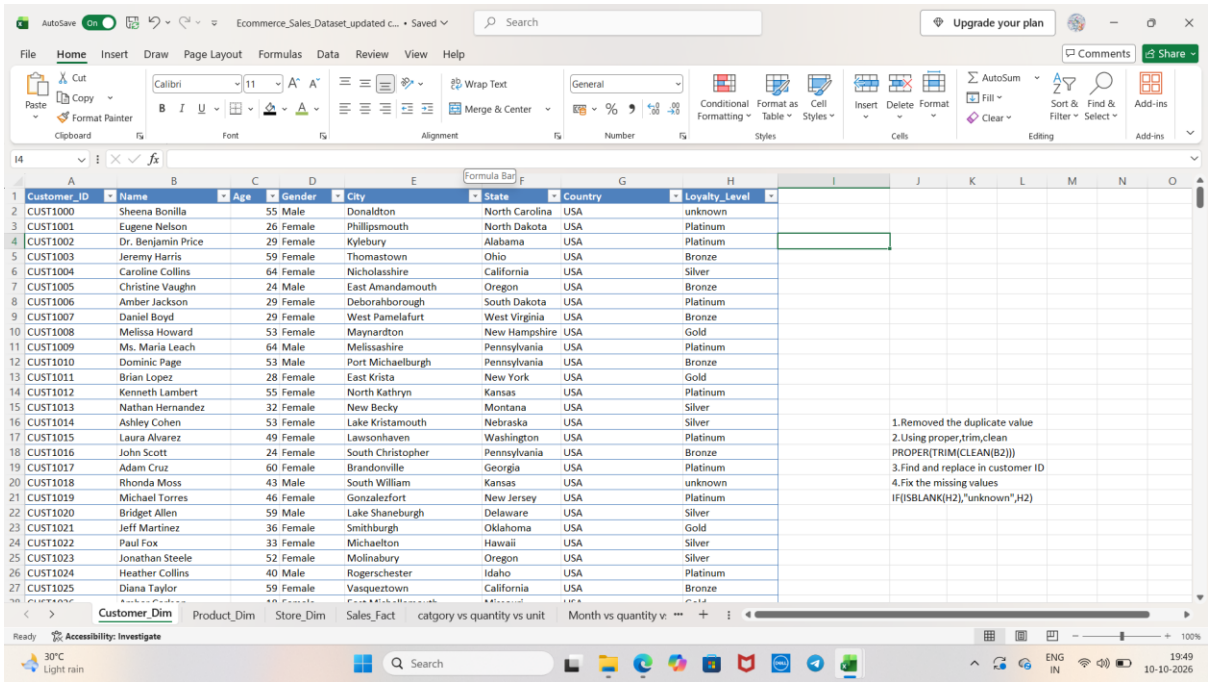
COLUMN DESCRIPTION:

COLUMN NAME	NAME OF THE COLUMN
COLUMN ID	CUSTOMER IDENTIFICATION
AGE	AGE OF THE CUSTOMERS
GENDER	TYPES OF GENDER
COUNTRY	COUNTRY OF THE CUSTOMER

PRODUCT ID	PRODUCT IDENTIFICATION
PRODUCT NAME	NAME OF THE PRODUCT
LOYALTY LEVEL	LOYALTY LEVEL OF CUSTOMER
CATEGORY	CATEGORIES OF THE PRODUCT
STOCK	STOCK LEVEL OF THE PRODUCT
STOCK ID	STOCK IDENTIFICATION
COST	COST OF THE PRODUCT
BRAND	BRAND OF THE PRODUCT
SALES ID	SALES IDENTIFICATION
ORDER DATE	DATE OF THE ORDER
QUANTITY	QUANTITY OF THE PRODUCT
UNIT PRICE	UNIT PRICE OF THE PRODUCT
TOTAL AMOUNT	TOTAL AMOUNT OF THE PRODUCT

DATA CLEANING ,TRANSFORMATION AND INSIGHTS:

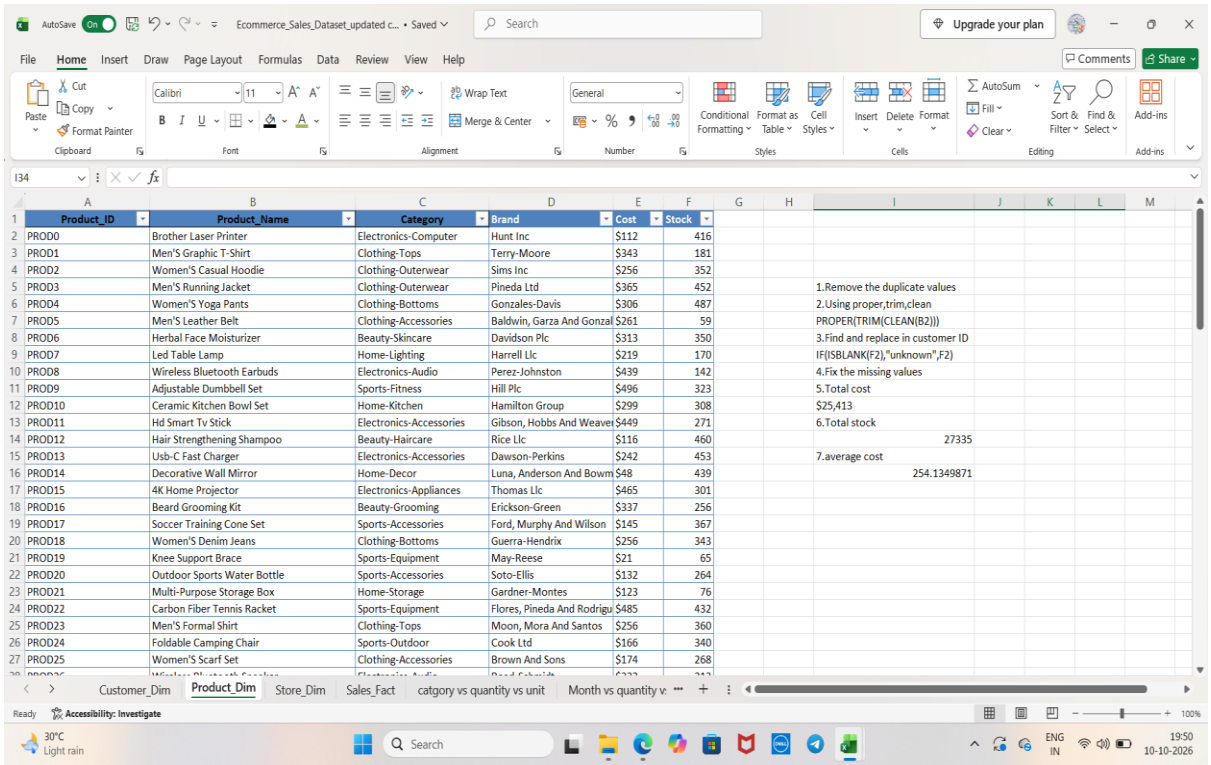
1.CUSTOMER _DIM:



Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	unknown
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	unknown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze

1.Removed the duplicate value
2.Using proper,trim,clean
PROPER(TRIM(CLEAN(B2)))
3.Find and replace in customer ID
4.Fix the missing values
IF(ISBLANK(H2),"unknown",H2)

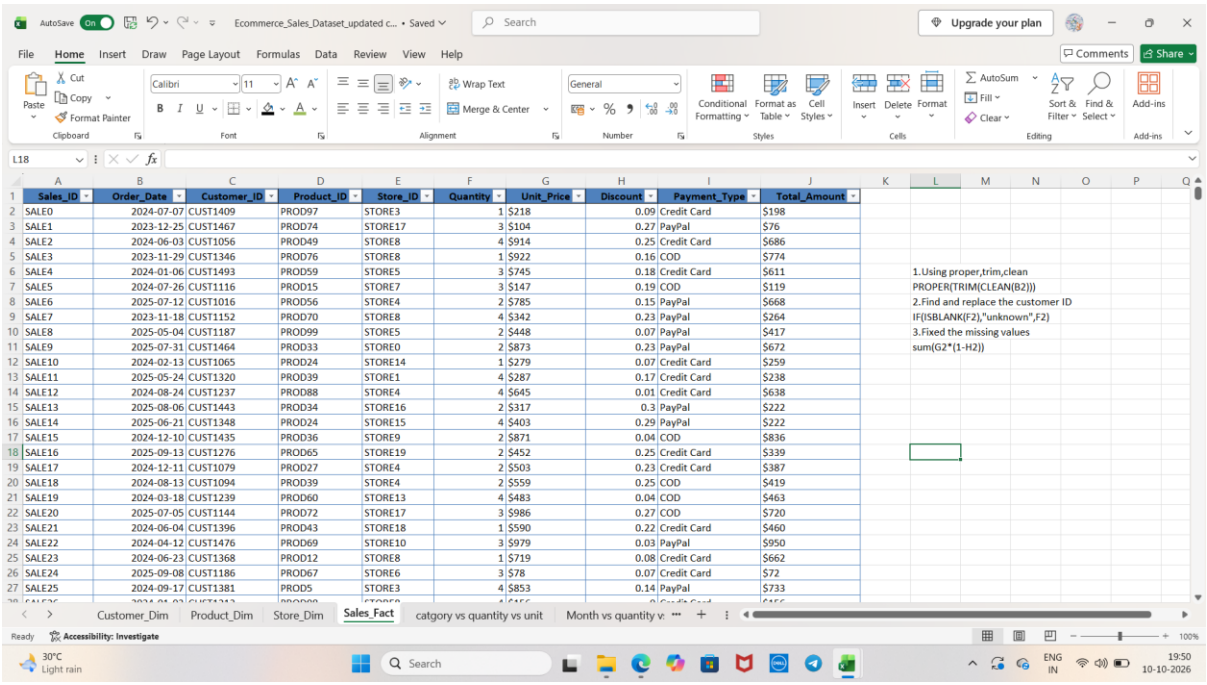
2.PRODUCT_DIM:



Product_ID	Product_Name	Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics-Computer	Hunt Inc	$112	416
PROD1	Men'S Graphic T-Shirt	Clothing-Tops	Terry-Moore	$343	181
PROD2	Women'S Casual Hoodie	Clothing-Outerwear	Sims Inc	$256	352
PROD3	Men'S Running Jacket	Clothing-Outerwear	Pineda Ltd	$365	452
PROD4	Women'S Yoga Pants	Clothing-Bottoms	Gonzales-Davis	$306	487
PROD5	Men'S Leather Belt	Clothing-Accessories	Baldwin, Garza And Gonzal	$261	59
PROD6	Herbal Face Moisturizer	Beauty-Skincare	Davidson Plc	$313	350
PROD7	Led Table Lamp	Home-Lighting	Harrell Llc	$219	170
PROD8	Wireless Bluetooth Earbuds	Electronics-Audio	Perez-Johnston	$439	142
PROD9	Adjustable Dumbbell Set	Sports-Fitness	Hill Plc	$496	323
PROD10	Ceramic Kitchen Bowl Set	Home-Kitchen	Hamilton Group	$299	308
PROD11	Hd Smart Tv Stick	Electronics-Accessories	Gibson, Hobbs And Weaver	$449	271
PROD12	Hair Strengthening Shampoo	Beauty-Haircare	Rice Llc	$116	460
PROD13	Usb-C Fast Charger	Electronics-Accessories	Dawson-Perkins	$242	453
PROD14	Decorative Wall Mirror	Home-Decor	Luna, Anderson And Bowm	$48	439
PROD15	4K Home Projector	Electronics-Appliances	Thomas Llc	$465	301
PROD16	Beard Grooming Kit	Beauty-Grooming	Erickson-Green	$337	256
PROD17	Soccer Training Cone Set	Sports-Accessories	Ford, Murphy And Wilson	$145	367
PROD18	Women'S Denim Jeans	Clothing-Bottoms	Guerra-Hendrix	$256	343
PROD19	Knee Support Brace	Sports-Equipment	May-Reese	$21	65
PROD20	Outdoor Sports Water Bottle	Sports-Accessories	Soto-Ellis	$132	264
PROD21	Multi-Purpose Storage Box	Home-Storage	Gardner-Montes	$123	76
PROD22	Carbon Fiber Tennis Racket	Sports-Equipment	Flores, Pineda And Rodrigu	$485	432
PROD23	Men'S Formal Shirt	Clothing-Tops	Moon, Mora And Santos	$256	360
PROD24	Foldable Camping Chair	Sports-Outdoor	Cook Ltd	$166	340
PROD25	Women'S Scarf Set	Clothing-Accessories	Brown And Sons	$174	268

1.Remove the duplicate values
2.Using proper,trim,clean
PROPER(TRIM(CLEAN(B2)))
3.Find and replace in customer ID
IF(ISBLANK(F2),"unknown",F2)
4.Fix the missing values
5.Total cost
$25,413
6.Total stock
27335
7.average cost
254.1349871

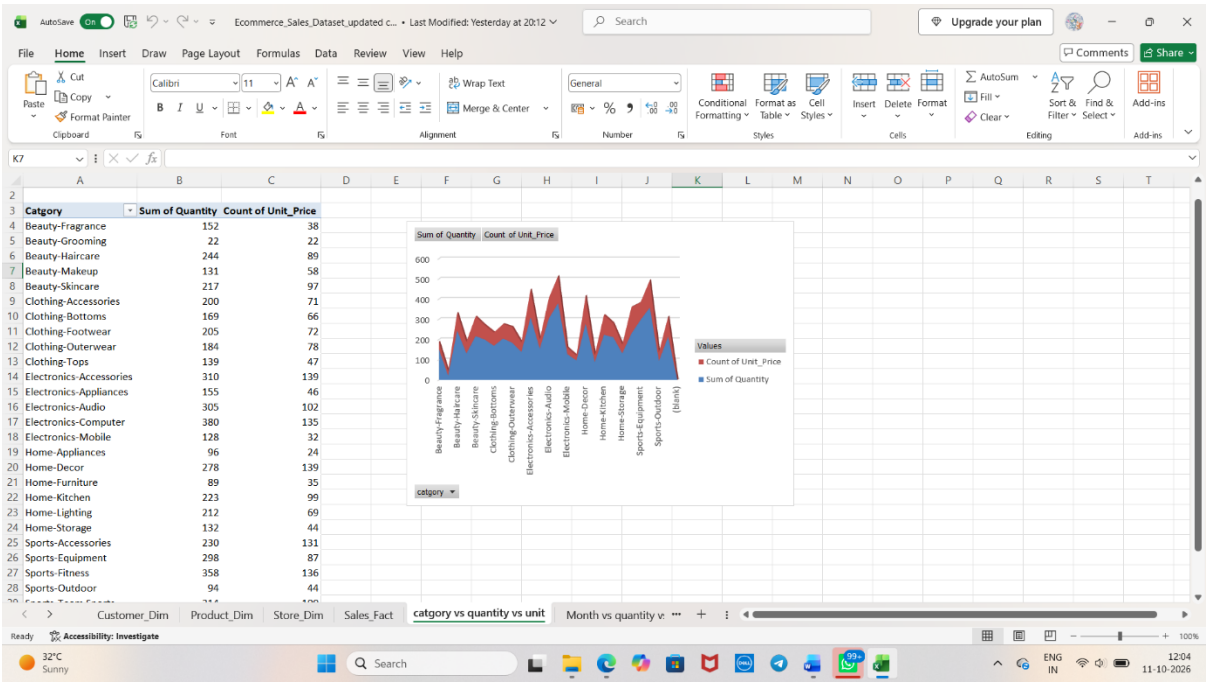
3.SALES_FACT:



Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	$218	0.09	Credit Card	$198
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	$104	0.27	PayPal	$76
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	$914	0.25	Credit Card	$686
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	$922	0.16	COD	$774
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	$745	0.18	Credit Card	$611
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	$147	0.19	COD	$119
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	$785	0.15	PayPal	$668
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	$342	0.23	PayPal	$264
SALE8	2025-05-04	CUST1187	PROD99	STORE5	2	$448	0.07	PayPal	$417
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	$873	0.23	PayPal	$672
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	$279	0.07	Credit Card	$259
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	$287	0.17	Credit Card	$238
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	$645	0.01	Credit Card	$638
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	$317	0.3	PayPal	$222
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	$403	0.29	PayPal	$222
SALE15	2024-12-10	CUST1435	PROD36	STORE9	2	$871	0.04	COD	$836
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	$452	0.25	Credit Card	$339
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	$503	0.23	Credit Card	$387
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	$559	0.25	COD	$419
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	$483	0.04	COD	$463
SALE20	2025-07-05	CUST1144	PROD72	STORE17	3	$986	0.27	COD	$720
SALE21	2024-06-04	CUST1396	PROD43	STORE18	1	$590	0.22	Credit Card	$460
SALE22	2024-04-12	CUST1476	PROD69	STORE10	3	$979	0.03	PayPal	$950
SALE23	2024-06-23	CUST1368	PROD12	STORE8	1	$719	0.08	Credit Card	$662
SALE24	2025-09-08	CUST1186	PROD67	STORE6	3	$78	0.07	Credit Card	$72
SALE25	2024-09-17	CUST1381	PROD5	STORE3	4	$853	0.14	PayPal	$733

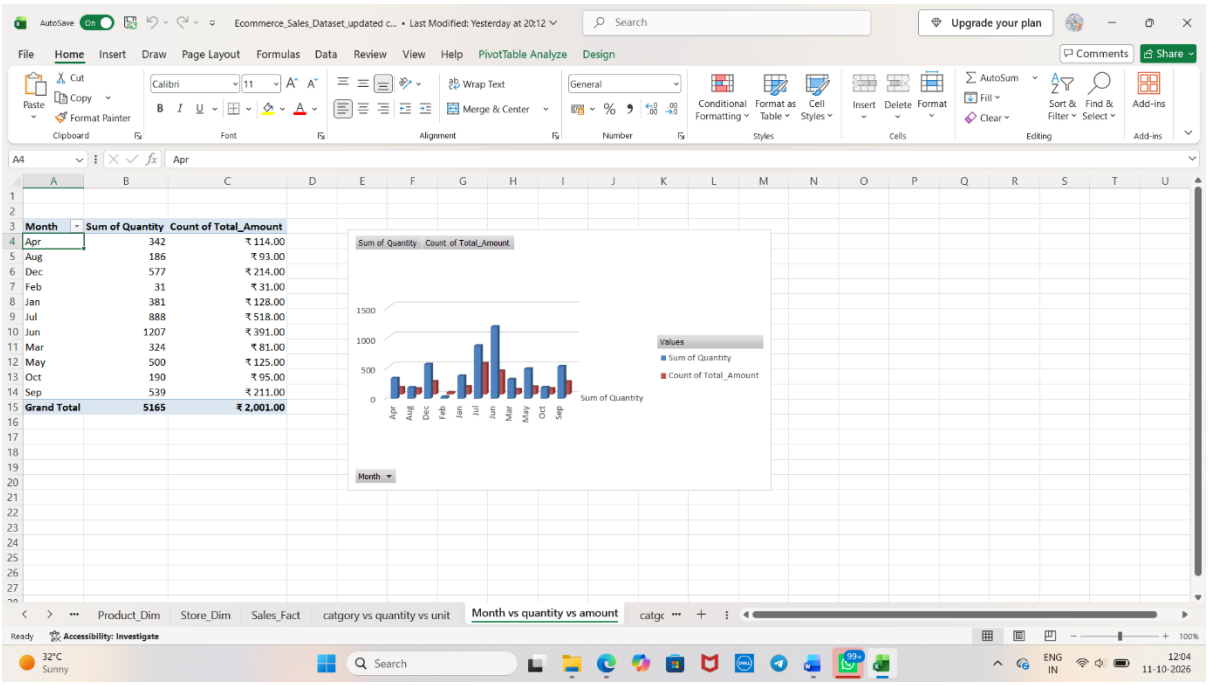
1.Using proper,trim,clean
PROPER(TRIM(CLEAN(B2)))
2.Find and replace the customer ID
IF(ISBLANK(F2),"unknown",F2)
3.Fixed the missing values
sum(G2*(1-H2))

4.CATEGORY VS QUANTITY VS UNIT



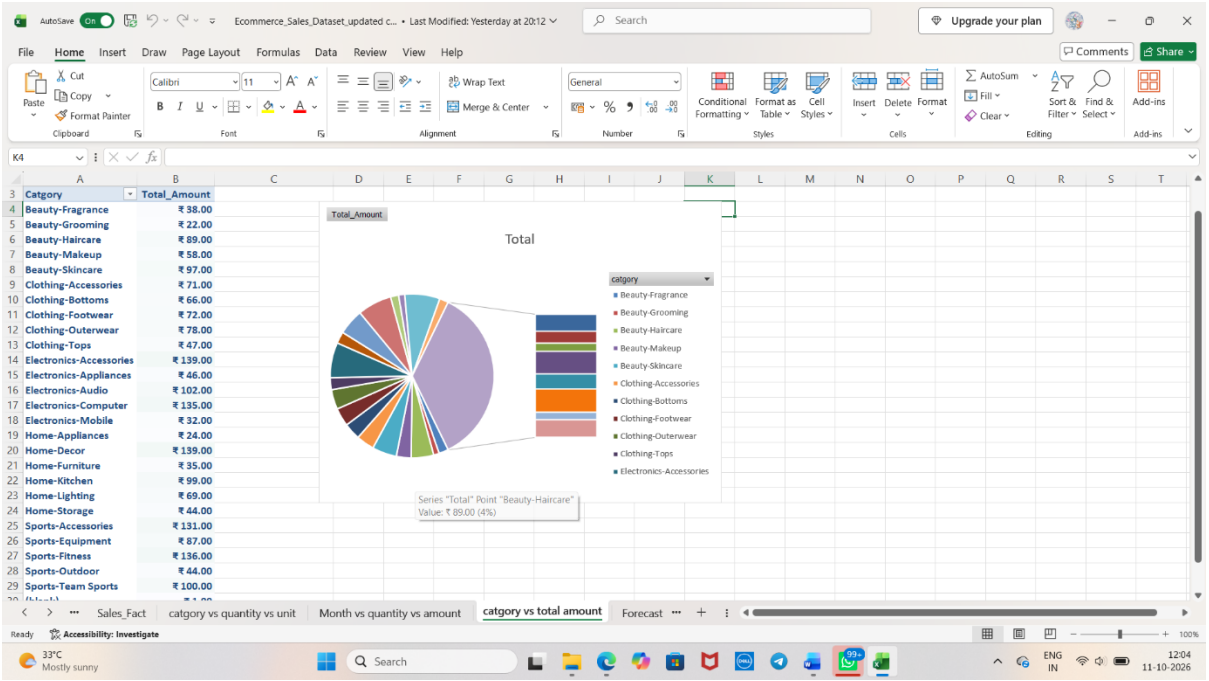
Catgory	Sum of Quantity	Count of Unit_Price
Beauty-Fragrance	152	38
Beauty-Grooming	22	22
Beauty-Haircare	244	89
Beauty-Makeup	131	58
Beauty-Skincare	217	97
Clothing-Accessories	200	71
Clothing-Bottoms	169	66
Clothing-Footwear	205	72
Clothing-Outerwear	184	78
Clothing-Tops	139	47
Electronics-Accessories	310	139
Electronics-Appliances	155	46
Electronics-Audio	305	102
Electronics-Computer	380	135
Electronics-Mobile	128	32
Home-Appliances	96	24
Home-Decor	278	139
Home-Furniture	89	35
Home-Kitchen	223	99
Home-Lighting	212	69
Home-Storage	132	44
Sports-Accessories	230	131
Sports-Equipment	298	87
Sports-Fitness	358	136
Sports-Outdoor	94	44

5.MONTH VS QUANTITY VS AMOUNT:



Month	Sum of Quantity	Count of Total_Amount
Apr	342	₹114.00
Aug	186	₹93.00
Dec	577	₹214.00
Feb	31	₹31.00
Jan	381	₹128.00
Jul	888	₹518.00
Jun	1207	₹391.00
Mar	324	₹81.00
May	500	₹125.00
Oct	190	₹95.00
Sep	539	₹211.00
Grand Total	**5165**	**₹2,001.00**

6.CATEGORY VS TOTAL AMOUNT



Catgory	Total_Amount
Beauty-Fragrance	₹38.00
Beauty-Grooming	₹22.00
Beauty-Haircare	₹89.00
Beauty-Makeup	₹58.00
Beauty-Skincare	₹97.00
Clothing-Accessories	₹71.00
Clothing-Bottoms	₹66.00
Clothing-Footwear	₹72.00
Clothing-Outerwear	₹78.00
Clothing-Tops	₹47.00
Electronics-Accessories	₹139.00
Electronics-Appliances	₹46.00
Electronics-Audio	₹102.00
Electronics-Computer	₹135.00
Electronics-Mobile	₹32.00
Home-Appliances	₹24.00
Home-Decor	₹139.00
Home-Furniture	₹35.00
Home-Kitchen	₹99.00
Home-Lighting	₹69.00
Home-Storage	₹44.00
Sports-Accessories	₹131.00
Sports-Equipment	₹87.00
Sports-Fitness	₹136.00
Sports-Outdoor	₹44.00
Sports-Team Sports	₹100.00

- The following steps all are used for data cleaning and transformation
  - Use Clean,Trim and Proper function to modify the Name Column.
  - Use Find and Replace method to modify United States of America into USA.
  - In Loyalty Level I can found some missing values use IF(ISBLANK([@[Loyalty_Level]]),"Unknown",[@[Loyalty_Level]]) Formula to fill the missing values in all the missing rows.
  - Use find and replace method to modify the product id's
  - No duplicate values are found.
  - In Brand column found some inconsistent data, I can use Clean,Trim and Proper function to modify all the inconsistent datas.

REPORT AND DASHBOARD:

- **Descriptive analysis:**

Use pivot table to analyse total sales, Average price and total cost.

- **Diagnostic Analysis:**

Use pivot table to analyse the which gender and age

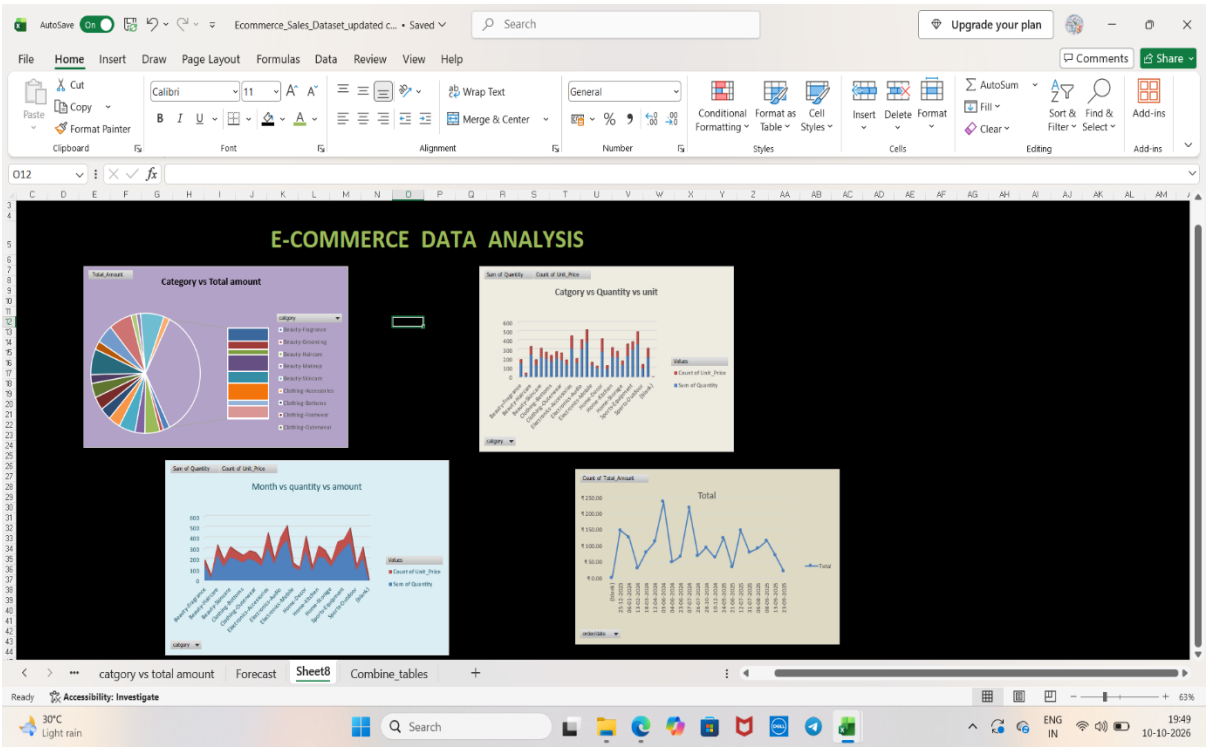
- **Predictive Analysis:**

Predictive analysis is based on line based price The highest value is upper line The lowest value is Lower line

- **Prescriptive Analysis:**

We can suggest more marketing strategies and discount strategies for low performing products

DASHBOARD:


E-COMMERCE DATA ANALYSIS
Category vs Total amount
Catgory vs Quantity vs unit
Month vs quantity vs amount
Total
catgory vs total amount
Forecast
Sheet8
Combine_tables